

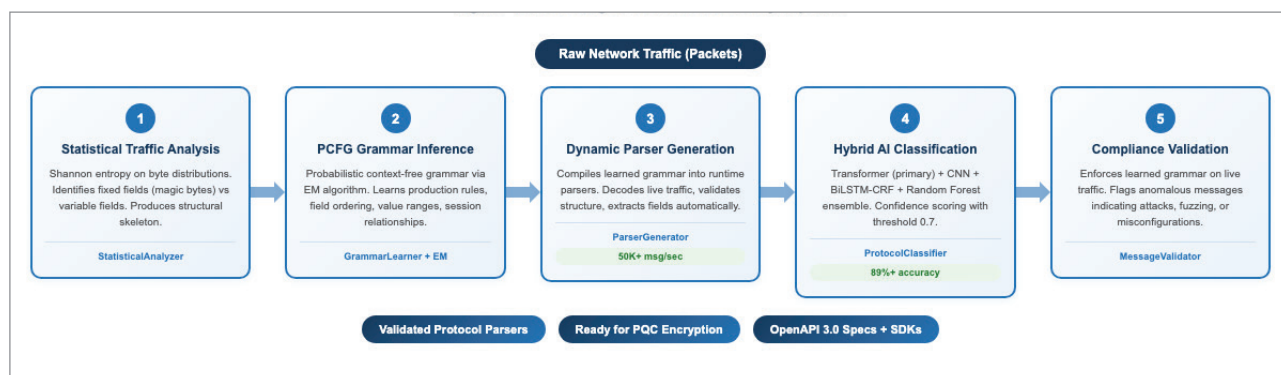


Figure 1: Five-stage protocol discovery pipeline

tools maintain signature databases. If a protocol is not in the database, it is invisible. QBITEL takes a fundamentally different approach: it learns protocol grammars from raw network traffic.

The discovery pipeline has five stages. Figure 1 illustrates how they connect.

Statistical traffic analysis: The *StatisticalAnalyzer* examines raw packet captures and computes field-level entropy, byte distribution patterns, and structural boundaries. Fixed-value fields (like protocol magic bytes or version numbers) are distinguished from variable-content fields (like user names or transaction amounts) using Shannon entropy calculations. This produces a structural skeleton of the unknown protocol.

Grammar learning with PCFG inference: The *GrammarLearner* uses a probabilistic context-free grammar (PCFG) inference engine refined with the expectation-maximisation (EM) algorithm. Starting from the structural skeleton, it learns production rules — which fields appear in which order, what the valid value ranges are, and how messages relate to each other across a session.

This is the same class of technique used in natural language processing to learn the grammar of a human language from a corpus of sentences — applied here to binary network protocols.

Dynamic parser generation: Once a grammar is learned, the *ParserGenerator* compiles it into a working parser at runtime. The generated parser can decode live traffic, validate message structure, and extract field values — all without any human writing parser code. The parser generation layer achieves 50,000+ messages per second throughput.

The hybrid AI classifier: The *ProtocolClassifier* uses a hybrid ensemble with the Transformer model as primary:

- **Transformer:** Multi-head self-attention for capturing long-range dependencies across protocol sessions — the primary classification engine.
- **CNN (convolutional neural network):** Detects local byte patterns — equivalent to finding words in the protocol vocabulary.

- **BiLSTM (bidirectional long short-term memory):** Models sequence structure with CRF for field-level detection.
- **Random forest:** Provides a robust statistical baseline for edge cases.

The ensemble vote produces a confidence score. Protocols above a configurable threshold (default 0.7) are promoted from candidate to known status and added to the protection perimeter.

Protocol compliance validation: The *MessageValidator* enforces the learned grammar against live traffic, flagging anomalous messages that could indicate attacks, protocol fuzzing, or misconfigured clients.

Real-world performance on first pass: 89%+ discovery accuracy, completing in 2 to 4 hours on a typical enterprise network segment, reducing what traditionally takes 6 to 12 months of manual reverse-engineering.

The post-quantum cryptography stack

Once protocols are discovered, QBITEL wraps them in post-quantum cryptography. The platform implements the full NIST post-quantum standardisation suite at Level 5, published as FIPS 203, 204, and 205 in 2024.

The PQCEngine class provides a single, domain-aware interface to all algorithms. Rather than requiring developers to select the right algorithm for their context, the engine accepts a *DomainProfile* value:

```
from ai_engine.crypto.pqc_unified import PQCEngine,
DomainProfile
```

```
# Healthcare: constrained devices (64 KB RAM), uses ML-
KEM-512
```

```
engine = PQCEngine(DomainProfile.HEALTHCARE)
```

```
# Automotive V2X: real-time (<1ms), uses Falcon for compact
signatures
```

```
engine = PQCEngine(DomainProfile.AUTOMOTIVE)
```